

Beyond Controlled-Domain Accuracy: Leakage-Aware Crop Disease Classification with Calibration and Multilingual Decision Support

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Abstract—Crop disease classification systems trained on controlled laboratory datasets frequently report near-perfect accuracy yet falter when exposed to field-acquired imagery. This paper presents a reproducible, leakage-aware pipeline for 19-class crop disease recognition derived from a curated PlantVillage subset spanning four crops (bell pepper, corn, potato, tomato). The pipeline combines near-duplicate auditing, frozen three-way data partitioning, convolutional and transformer classifiers, post-hoc temperature-scaling calibration, confidence-based selective prediction, a heuristic visible lesion-area proxy, zero-shot external evaluation on PlantDoc, and a retrieval-grounded multilingual advisory prototype.

EfficientNet-B0, pre-specified as the baseline model, achieves 99.53% top-1 accuracy and 0.9929 macro-F1 on the locked 5,322-image internal test set. Temperature scaling ($T = 1.5695$) reduces expected calibration error (ECE) from 0.0032 to 0.0023 and Brier score from 0.00704 to 0.00642, while internal-test negative log-likelihood increases slightly from 0.0155 to 0.0159; calibration improvement is therefore metric-dependent rather than uniform. An EfficientNetV2-S proposed model and a ViT-B/16 comparator reach comparable internal accuracy but offer no material advantage under this evaluation protocol.

Zero-shot evaluation on a 144-image, 15-class semantically mapped PlantDoc subset reveals sharp domain shift: top-1 accuracy drops to 19.44% while top-5 accuracy remains 67.36%, indicating that the correct class is retrievable within the model’s top predictions but is not reliably top-ranked under field-like conditions. Strong controlled-domain performance did not transfer

reliably to the semantically mapped PlantDoc external subset, demonstrating that high internal accuracy alone is insufficient evidence of robustness to field-like imaging conditions.

Index Terms—crop disease detection, EfficientNet, Vision Transformer, temperature scaling, calibration, selective prediction, domain shift, PlantDoc, retrieval-augmented generation, multilingual advisory, PlantVillage

I. INTRODUCTION

Crop diseases are responsible for substantial yield losses worldwide, with estimates from the Food and Agriculture Organization of the United Nations (FAO) suggesting that plant pathogens and pests account for up to 40% of annual global crop production losses [1]. For smallholder farmers with limited access to extension services, early and accurate disease identification is a critical determinant of treatment timing and economic outcome [2]. Automated computer-vision systems trained on large annotated datasets offer the potential to democratise field-level disease diagnosis [3], [4], particularly when deployed on mobile platforms accessible in low-connectivity regions [2], [5].

The PlantVillage dataset [6], comprising over 54,000 images collected under controlled, single-leaf conditions against uniform backgrounds, has become the dominant benchmark for crop disease classification. Models trained on PlantVillage routinely achieve accuracies exceeding 95% [3], [7], [8]. However, a growing body of evidence cautions against interpreting such figures as deployment readiness. Controlled-domain datasets exhibit a fundamental mismatch with the com-

*Mentored the research work.

plex backgrounds, variable lighting, overlapping leaves, and partial occlusions encountered in real agricultural fields [4], [9]. This domain shift problem has been demonstrated empirically through the PlantDoc dataset [10], which provides field-acquired images and enables direct measurement of the gap between controlled and field-like performance.

Beyond raw accuracy, well-calibrated confidence estimates are essential for safe model deployment. Overconfident predictions—wherein the model assigns high softmax probabilities to incorrect classes—are particularly hazardous in high-stakes agricultural decisions [11]. Temperature scaling [11] is a widely used post-hoc calibration method that rescales logits by a learned scalar, improving confidence alignment without altering top-1 predictions. Selective prediction frameworks [12] complement calibration by allowing the model to abstain on low-confidence inputs, providing a mechanism for reducing error rates at the cost of partial coverage.

Finally, even when a reliable visual classifier is available, farmers require actionable guidance in their native language. Retrieval-augmented generation (RAG) systems [13] can serve as a structured advisory interface that grounds responses in vetted agronomic knowledge bases while enforcing safety constraints against prescriptive pesticide recommendations.

A. Research Objectives

This study addresses the following objectives:

- 1) Construct a reproducible, leakage-reduced 19-class crop disease classification pipeline on PlantVillage-derived data using strict partition isolation.
- 2) Evaluate and compare EfficientNet-B0, EfficientNetV2-S, and ViT-B/16 classifiers on controlled-domain internal metrics and selective prediction behaviour.
- 3) Quantify zero-shot domain shift by evaluating the pre-specified baseline model on the external PlantDoc test set under a transparent semantic mapping protocol.
- 4) Introduce a heuristic visible lesion-area proxy with explicitly stated limitations as a qualitative image-characterisation tool.
- 5) Design and functionally test a safety-bounded multilingual retrieval-grounded advisory prototype that conditions guidance on classifier confidence bands.

B. Contributions

The principal contributions are:

- A reproducible, near-duplicate-audited 19-class data split protocol with documented seed, class mapping, and partition counts, including transparent reporting of reviewed and unreviewed near-duplicate candidates.
- Multi-architecture benchmarking on the same frozen partitions, showing that under the evaluated models and split protocol, larger architectures did not show a practically meaningful internal-domain advantage over the compact EfficientNet-B0 baseline.

- A transparent zero-shot PlantDoc evaluation of the proposed pipeline, quantifying the controlled-to-field performance gap.
- A qualitative heuristic visible lesion-area proxy with documented protocol limitations, explicitly precluding agronomic or clinical severity claims.
- A prototype multilingual retrieval-grounded advisory layer functionally tested across all 19 labels in English, Hindi, and Telugu, with a tiered confidence-safety policy.

II. RELATED WORK

A. Plant Disease Datasets and Benchmarking

Hughes and Salathé [6] introduced PlantVillage as the first large-scale open-access leaf-disease image repository. Subsequent work has critiqued the controlled capture conditions as unrepresentative of field settings [3], [9]. Singh et al. [10] addressed this gap with PlantDoc, a 2,600-image multi-crop dataset collected under natural field conditions. Additional datasets targeting specific crops, including cassava [2], [14], rice, and tomato [15] have been developed, but PlantVillage remains the standard multi-crop controlled benchmark. Dataset governance issues such as duplicate images across splits have been identified as a source of inflated accuracy estimates, motivating careful de-duplication [16], [17].

B. CNN-Based Plant Disease Classification

Mohanty et al. [3] pioneered the application of deep CNNs to PlantVillage, achieving 99.35% accuracy under colour-augmented training. Ferentinos [4] reported similarly high accuracy using AlexNet, VGG, and GoogLeNet variants. The effect of transfer learning from ImageNet-pretrained weights has been consistently observed to accelerate convergence and improve final accuracy [7], [8]. Data augmentation strategies including random cropping, horizontal flipping, brightness jitter, and colour-space transformations are standard practice.

C. Efficient Neural Architectures

Tan and Le introduced EfficientNet [18], which applies compound scaling to simultaneously balance network width, depth, and input resolution. The successor EfficientNetV2 [19] further improved training speed through progressive learning and Fused-MBConv blocks. Applications in agricultural disease recognition have demonstrated that EfficientNet variants offer a favourable accuracy-efficiency tradeoff [5].

D. Vision Transformers in Agricultural Applications

Dosovitskiy et al. [20] introduced the Vision Transformer (ViT) that processes image patches as token sequences with self-attention. Data Efficient Image Transformers (DeiT) addressed ViT's large data requirement through distillation. ViT and Swin Transformer have been applied to plant disease recognition, sometimes matching or exceeding CNN performance on benchmark datasets. However, transformer architectures typically require more parameters per model, limiting their practical advantage in resource-constrained deployments.

E. Probability Calibration and Selective Prediction

Calibration refers to the agreement between a model’s predicted confidence and its empirical accuracy [11]. Temperature scaling divides all logits by a positive scalar T optimised to minimise negative log-likelihood on a held-out calibration set [11], [21]. Dividing all logits by a positive scalar preserves their ordering and therefore preserves the top-1 predicted class; it changes the softmax probability distribution and associated confidence metrics. Selective classification [12] introduces an abstention option: the model declines to classify inputs below a chosen confidence threshold, trading coverage for improved accuracy on the retained subset. Overconfidence under distribution shift has been highlighted as a critical failure mode for safety-critical applications [22].

F. Domain Adaptation and External Validation

The domain gap between controlled-laboratory and field-acquired data is a well-documented challenge in agricultural computer vision [9]. Approaches include domain adaptation, self-supervised pre-training, and test-time augmentation. However, direct zero-shot evaluation—applying a source-domain-trained model to target-domain data without any adaptation—remains the most conservative and honest estimate of model robustness under distribution shift. Several studies have demonstrated large performance drops when PlantVillage models are evaluated on field data [10], [15].

G. Explainability and Lesion Analysis

Grad-CAM [23] and SHAP provide model-specific and model-agnostic attribution, respectively. Lesion segmentation methods range from threshold-based colour separation to deep semantic segmentation. Severity estimation—the quantification of disease extent within a leaf—typically requires expert-annotated pixel masks for validation. In the absence of validated annotations, any automatic severity estimate must be characterised as a heuristic proxy.

H. Retrieval-Augmented Advisory Systems

RAG systems [13] combine a retrieval module over a curated knowledge base with a language model to generate grounded responses. Responsible deployment of AI advisory systems in agriculture requires explicit safety constraints, uncertainty communication, and avoidance of unverified pesticide or chemical recommendations. Multilingual capability is particularly important for serving smallholder farmers in South Asia.

III. MATERIALS AND METHODS

A. Datasets, Class Definition, and Data Governance

The primary dataset is a curated subset of PlantVillage [6], selecting five economically important crop species: bell pepper (*Capsicum annuum*), corn (*Zea mays*), potato (*Solanum tuberosum*), soybean (*Glycine max*), and tomato (*Solanum lycopersicum*). The initial selected-source pool contained **31,729** images across 20 crop–disease label combinations. Five images identified as exact validation-set duplicates were removed

TABLE I
MAIN19 DATASET FLOW. EACH ROW REMOVES A SPECIFIC SET OF IMAGES. THE FINAL MAIN19 TOTAL IS THE SUM OF THE THREE LOCKED PARTITIONS.

Stage	Images
Initial selected-source pool (5 crops, 20 classes)	31,729
– Exact validation-set duplicates	–5
= EDA-clean pool	31,724
– Confirmed near-duplicate cross-split exclusions	–2
= Final locked pool (Aux20, 20 classes)	31,722
– Excluded class: <code>soybean__healthy</code>	–5,090
= Final Main19 total (19 classes, 4 crops)	26,632
Training split	18,113
Calibration split	3,197
Locked internal test split	5,322

during exploratory data analysis, leaving 31,724 clean images across 20 classes and 5 crops (Table I).

The primary classification task, designated **Main19**, excludes the `soybean__healthy` class, which was reserved for a secondary auxiliary experiment (Aux20). After excluding `soybean__healthy` and removing two confirmed near-duplicate cross-split images (see Section III-B), the final Main19 task contains **19 classes** spanning **four crops**: bell pepper, corn, potato, and tomato. The 19 labels comprise 4 healthy-state classes and 15 diseased-state classes. The largest class (tomato yellow leaf curl virus) contains 5,357 images; the smallest (potato healthy) contains 152 images, yielding a class imbalance ratio of 35.24.

The external evaluation dataset is PlantDoc [10], a field-acquired, multi-crop leaf-disease dataset. Only the PlantDoc test split was used; no PlantDoc image was accessed for training, model selection, temperature fitting, or threshold calibration, ensuring that the PlantDoc evaluation is strictly zero-shot.

B. Duplicate Auditing and Reproducible Data Splits

Near-duplicate detection was conducted using perceptual hashing (pHash) with a Hamming-distance threshold of ≤ 4 bits, yielding **136** near-duplicate candidate pairs [24]. Of these, **2** were confirmed as true cross-split near-duplicates and excluded from all downstream partitions. The remaining **134** candidate pairs were flagged for review but remain **unreviewed** in the repository audit log. All 136 candidates are cross-split pairs (one image from the original training pool and one from the original validation set).

Because the review was not exhaustive, we do not claim that the final split is completely free of near-duplicate leakage. Rather, the audit reduced documented leakage risk by identifying and removing the two highest-similarity confirmed duplicates. The 134 unreviewed candidates represent a residual risk that could be addressed by future manual or automated review.

Reproducible three-way stratified splits were constructed at random seed 7. The original PlantVillage training pool served

as the development pool from which 15% were stratified-sampled to form the calibration split. The original PlantVillage validation set was retained unchanged as the locked internal test set (after the two near-duplicate exclusions). Partition membership was frozen and saved as versioned manifests before any model training commenced.

The final locked Main19 partition counts are: **training**: 18,113 images; **calibration**: 3,197 images; **internal test**: 5,322 images. Fig. 1 shows the per-class image counts across all three partitions.

C. Image Classification Models and Training Protocol

Three architectures were evaluated on the Main19 task. All models were initialised with ImageNet-pretrained weights (IMAGENET1K_V1) and fine-tuned on the training split. Checkpoint selection for each model was governed exclusively by calibration-set macro-F1; the internal test set was not accessed during training or checkpoint selection for any of the three models. The internal test set was accessed only once, for final one-shot evaluation after all checkpoints had been frozen.

EfficientNet-B0 (Baseline). The baseline classifier, pre-specified as the primary model for downstream calibration, external evaluation, and advisory integration, uses EfficientNet-B0 [18] with a 19-class linear head. Images were resized to 224×224 pixels. Training used class-weighted cross-entropy loss with AdamW [25] at learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 32, and a maximum of 15 epochs with early stopping after 4 non-improving calibration epochs. Step-decay halved the learning rate at epochs 6 and 13. The final checkpoint was selected at epoch 15 (calibration macro-F1: 0.9940). Notebook metadata confirms `internal_test_used: false`.

EfficientNetV2-S (Proposed). The proposed model replaces the backbone with EfficientNetV2-S [19]. Input resolution was increased to 384×384 pixels. Training used class-weighted focal cross-entropy loss [26] ($\gamma = 1.5$) with AdamW at learning rate 2×10^{-4} , weight decay 10^{-4} , batch size 12, and a maximum of 20 epochs (early stopping patience: 5). The best checkpoint was selected at epoch 13 (calibration macro-F1: 0.9979). Notebook metadata confirms `internal_test_used: false`.

ViT-B/16 (Architecture Comparator). Vision Transformer ViT-B/16 [20] was trained as an architecture comparator. Input size was 224×224 . Training used class-weighted cross-entropy with label smoothing ($\varepsilon = 0.05$) [27] and AdamW at learning rate 10^{-4} , weight decay 10^{-4} , batch size 16, and a maximum of 15 epochs (early stopping patience: 4). The best checkpoint was selected at epoch 11 (calibration macro-F1: 0.9947). Notebook metadata confirms `internal_test_used_during_training: false`.

All experiments were conducted in PyTorch 2.10.0+cu128, Python 3.12.13, NumPy 2.0.2, and pandas 2.3.3 at seed 7. Training was performed on NVIDIA GPU hardware via the Kaggle platform. Table II summarises the model configurations.

D. Post-hoc Calibration

Following Guo et al. [11], temperature scaling was applied to the EfficientNet-B0 baseline by fitting a scalar temperature parameter T that minimises negative log-likelihood on the calibration split. The optimal temperature was $T = 1.5695$. Because dividing all logits by a positive scalar preserves their ordering, the top-1 class assignment is invariant to temperature scaling: accuracy and macro-F1 are identical before and after scaling. The transformation changes the softmax probability distribution and therefore affects confidence-dependent metrics (ECE, Brier score, NLL).

Expected calibration error (ECE) [28] was computed with 15 equal-width bins. Brier score [29] was computed as the mean squared error between predicted probability vectors and one-hot targets, averaged over samples: $\text{Brier} = \frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C (p_{ic} - y_{ic})^2$.

Temperature-scaled internal-test metrics were extracted only for EfficientNet-B0 (Notebook 04). Independent temperature-scaled internal-test evaluation for EfficientNetV2-S was not conducted; accordingly, no temperature-scaled test comparison across models is reported.

E. Selective Prediction

Selective prediction was implemented for all three architectures following the risk-coverage framework of Geifman and El-Yaniv [12]. A confidence threshold τ was selected on the calibration split under the constraint that the calibration-set error rate (risk) was $\leq 1\%$ while maximising coverage, with a minimum threshold floor of $\tau \geq 0.50$. Thresholds were selected exclusively on calibration data; the internal test set was not used for threshold tuning.

For all three models, the selected threshold was $\tau = 0.50$, which is the minimum floor. The confidence values used for selective prediction are the raw (uncalibrated) softmax outputs from each model’s respective checkpoint; temperature scaling was not applied before selective prediction thresholding.

F. Heuristic Visible Lesion-Area Proxy

A heuristic image-based visible lesion-area proxy was computed for all Main19 images (including healthy classes). The image-processing procedure estimates the proportion of visible pixels satisfying pre-defined lesion-like colour and texture criteria, applying background separation followed by HSV/Lab abnormal-colour and low-green vegetation heuristics. The method requires no separate lesion segmentation model and no expert annotation.

The proxy assigns each image to one of three ordinal bands: *mild_proxy* ($< 5\%$ visible lesion area), *moderate_proxy* ($5\text{--}20\%$), and *severe_proxy* ($> 20\%$). These labels are visual-heuristic designations only. The protocol metadata explicitly states that the proxy: (i) is not expert-annotated lesion segmentation; (ii) is not a clinical disease-stage label; (iii) has not been validated against agronomic treatment thresholds; and (iv) does not account for hidden lesions, leaf age, whole-plant burden, or temporal progression. All proxy values in this paper are accordingly qualified as heuristic estimates.

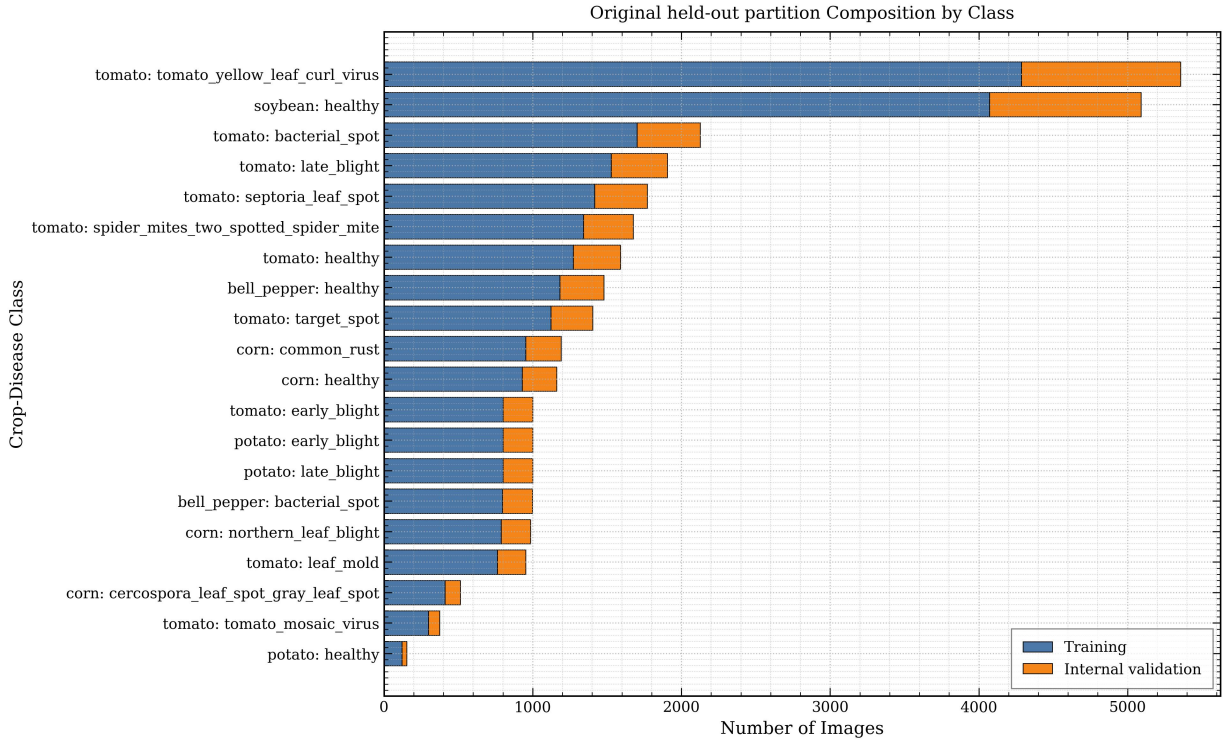


Fig. 1. Image counts per Main19 class across the three frozen partitions (training: 18,113; calibration: 3,197; internal test: 5,322). Classes are ordered by total size. All partitions were stratified by class and their membership frozen before any model training. The 19 classes span four crops: bell pepper (2), corn (4), potato (3), and tomato (10).

TABLE II

MODEL ARCHITECTURES AND TRAINING CONFIGURATIONS. WTD. CE DENOTES CLASS-WEIGHTED CROSS-ENTROPY; LS DENOTES LABEL SMOOTHING; ES DENOTES EARLY STOPPING. ALL CHECKPOINTS WERE SELECTED BY CALIBRATION-SET MACRO-F1. THE INTERNAL TEST SET WAS NOT USED FOR CHECKPOINT SELECTION IN ANY MODEL.

Property	EfficientNet-B0	EfficientNetV2-S	ViT-B/16
Input size	224×224	384×384	224×224
Parameters	≈ 5.3 M	≈ 20.2 M	85.8 M
Loss	Wtd. CE	Wtd. Focal CE	Wtd. CE + LS
Focal γ	—	1.5	—
Label smoothing	None	None	0.05
Optimiser	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Weight decay	1×10^{-4}	1×10^{-4}	1×10^{-4}
Batch size	32	12	16
Maximum epochs	15	20	15
ES patience	4	5	4
Selected epoch	15	13	11
Calibration macro-F1	0.9940	0.9979	0.9947

This procedure does not identify causal model features and does not constitute classifier-attention analysis or model-specific explainability. It is an exploratory image-characterisation measure.

G. Zero-Shot External-Domain Evaluation

To measure the practical impact of source-to-target domain shift, the EfficientNet-B0 baseline was applied zero-shot to

the PlantDoc test split. “Zero-shot” means no PlantDoc image was used for training, model selection, calibration, or threshold tuning. EfficientNet-B0 was pre-specified as the baseline model before internal-test evaluation; the decision to use it for external evaluation did not depend on internal-test results.

PlantDoc organises images into 27 folder categories. Of these, **15** could be *semantically mapped* to Main19 classes based on folder names and disease terminology. The re-

maintaining 12 folders cover crops absent from Main19 (apple, blueberry, cherry, grape, peach, raspberry, soybean, squash, strawberry) and were excluded. Two mappings rely on assumptions about generic folder names: `Tomato_leaf` was mapped to `tomato_healthy` and `Bell_pepper_leaf` to `bell_pepper_healthy`. These folder-to-healthy mappings are semantic assumptions that could affect class-level metrics if the folders contain non-healthy images.

Evaluation was conducted on the model’s unscaled softmax outputs (temperature $T = 1.0$). Temperature scaling fitted on PlantVillage calibration data was not applied to PlantDoc, because its application to a different data distribution is an unjustified extrapolation.

The model outputs predictions across all 19 Main19 classes, but only 15 of those classes have corresponding PlantDoc true labels. The confusion matrix therefore has 15 true-label rows; predictions into the remaining 4 Main19 classes (those without PlantDoc counterparts) are visible in the confusion matrix columns as off-diagonal entries.

H. Retrieval-Grounded Multilingual Advisory Prototype

The retrieval-grounded advisory prototype provides safety-bounded educational guidance following a predicted diagnosis. It is not an image classifier and does not alter classification outputs. The knowledge base contains 19 entries, one per Main19 label, with structured agronomic content including symptom descriptions, causal agents, management principles, and prevention guidelines. All entries were authored for this project and have not been reviewed or endorsed by professional plant pathologists.

Retrieval uses TF-IDF cosine similarity with classifier-label prioritisation [30]: when a classifier label is provided, that entry is retrieved directly. Because there is one entry per label, correct primary retrieval is deterministic rather than a competitive search over a large corpus. Responses are generated in three languages: English, Hindi, and Telugu.

The advisory layer enforces a tiered confidence-safety policy: *low confidence* (< 0.70): requests clearer images and expert confirmation; *medium confidence* ($0.70\text{--}0.89$): provides screening-level guidance and recommends inspecting multiple plants; *high confidence* (≥ 0.90): provides substantive agronomic guidance with mandatory field-verification warnings. Across all tiers, the system explicitly prohibits pesticide brand, dosage, or location-specific chemical recommendations.

IV. RESULTS

A. Dataset Characteristics and Split Integrity

The Main19 split yielded 18,113 training, 3,197 calibration, and 5,322 internal test images across 19 classes and 4 crops (Table I). Stratification preserved the class proportions across all three partitions (Fig. 1). Near-duplicate auditing confirmed 2 cross-split near-duplicate pairs from 136 pHash candidates. Both were excluded. The remaining 134 candidates are unreviewed.

B. Controlled-Domain Classification Performance

EfficientNet-B0 Baseline. The baseline model achieves **99.53%** top-1 accuracy, 99.32% balanced accuracy, and **0.9929** macro-F1 on the locked internal test set (source: Notebook 04 test evaluation). Fig. 2 shows the training and calibration loss and macro-F1 across 15 epochs. Training converged steadily; calibration macro-F1 reached 0.9940 at the final epoch, which was selected as the best checkpoint.

Multi-Architecture Comparison. A post-hoc descriptive comparison was conducted in Notebook 07, evaluating all three frozen checkpoints on the same locked internal test set. This comparison was descriptive only; the EfficientNet-B0 baseline had been pre-specified as the primary model before internal-test evaluation. Table III presents the results.

Cross-notebook metric discrepancy. The EfficientNet-B0 accuracy differs between Notebook 04 (0.9953025) and the Notebook 07 comparison table (0.9954904), a difference of one additional correct prediction out of 5,322. The Brier scores also differ substantially: Notebook 04 reports 0.00704, while Notebook 07 reports 0.000337. This 20-fold discrepancy indicates that the two notebooks use different Brier score computation implementations. We adopt the Notebook 04 values as canonical for EfficientNet-B0, because Notebook 04 is the dedicated temperature-scaling and test-evaluation notebook and its Brier formula is documented as the standard multiclass definition. The Brier values from the Notebook 07 comparison table are not directly comparable to Notebook 04 and are therefore not used for cross-model calibration claims.

Under the evaluated models and split protocol, larger architectures did not show a practically meaningful internal-domain advantage over EfficientNet-B0: all three models achieved accuracy above 99.3% and macro-F1 above 0.992, while EfficientNet-B0 has approximately $16\times$ fewer parameters than ViT-B/16.

C. Calibration and Confidence-Aware Prediction

Temperature Scaling. Temperature scaling with $T = 1.5695$ preserved accuracy (0.9953) and macro-F1 (0.9929) identically, reduced internal-test ECE from 0.00321 to 0.00228 (a 29% relative reduction), and reduced Brier score from 0.00704 to 0.00642. However, internal-test NLL *increased slightly* from 0.01550 to 0.01593. Calibration improvement was therefore metric-dependent rather than uniform. Table IV presents the full profile.

The reliability diagram (Fig. 3) confirms that the temperature-scaled model exhibits closer alignment between mean confidence bins and empirical accuracy on the internal test set.

Selective Prediction. For all three models, the calibration-selected threshold was $\tau = 0.50$ (the minimum floor). At this threshold, EfficientNet-B0 retained all 5,322 internal test images (coverage: 100%), achieving selective accuracy of 0.9955 and risk of 0.0045. EfficientNetV2-S rejected 2 images (coverage: 99.96%), ViT-B/16 rejected 3 (coverage: 99.94%). At the calibration-selected threshold, EfficientNet-B0 retained all internal-test images because predictions on the matched

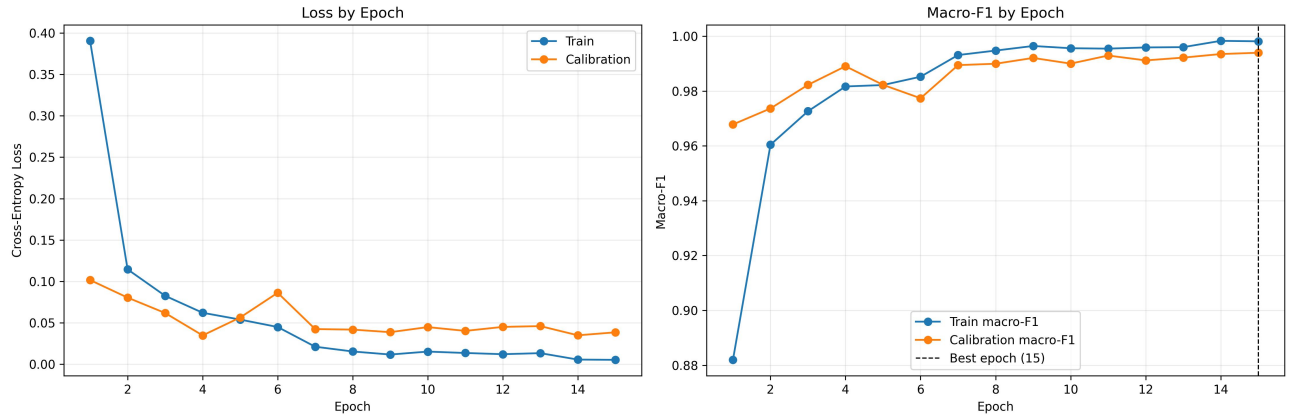


Fig. 2. EfficientNet-B0 optimisation on the Main19 development protocol. Left: training and calibration loss. Right: macro-F1 by epoch. The selected checkpoint corresponds to the highest calibration macro-F1 (0.9940 at epoch 15). The internal test partition was not used for epoch selection (18,113 training images, 3,197 calibration images).

TABLE III

CONTROLLED-DOMAIN PERFORMANCE ON THE MAIN19 INTERNAL TEST SET (5,322 IMAGES). HIGHER IS BETTER FOR ACCURACY, BALANCED ACCURACY, AND MACRO-F1; LOWER IS BETTER FOR NLL. B0 CANONICAL VALUES FROM THE DEDICATED EVALUATION (NOTEBOOK 04); V2-S AND ViT-B/16 FROM THE POST-HOC COMPARISON (NOTEBOOK 07). SEE TEXT FOR CROSS-NOTEBOOK DISCREPANCY DISCUSSION.

Model	Source	Accuracy	Balanced Acc.	Macro-F1	NLL
EfficientNet-B0	NB04 (canonical)	0.9953	0.9932	0.9929	0.0155
EfficientNet-B0	NB07 (comparison)	0.9955	0.9935	0.9931	0.0159
EfficientNetV2-S	NB07	0.9955	0.9935	0.9926	0.0176
ViT-B/16	NB07	0.9936	0.9938	0.9926	0.1481

TABLE IV

EFFICIENTNET-B0 CALIBRATION RESULTS BEFORE AND AFTER TEMPERATURE SCALING ($T = 1.5695$). ACCURACY AND MACRO-F1 ARE UNCHANGED BECAUSE POSITIVE LOGIT SCALING PRESERVES CLASS RANKING. LOWER VALUES ARE BETTER FOR NLL, BRIER SCORE, AND ECE. ECE COMPUTED WITH 15 EQUAL-WIDTH BINS. BRIER: MULTICLASS MEAN SQUARED ERROR, $\frac{1}{N} \sum_i \sum_c (p_{ic} - y_{ic})^2$.

Split	Scaling	Accuracy	NLL	Brier score	ECE
Calibration	Uncalibrated	0.9959	0.01850	0.00699	0.00392
	Temperature-scaled	0.9959	0.01830	0.00678	0.00130
Internal test	Uncalibrated	0.9953	0.01550	0.00704	0.00321
	Temperature-scaled	0.9953	0.01593	0.00642	0.00228

internal domain were already highly confident. Accordingly, the reject option produced no additional in-domain benefit for this model.

Fig. 4 shows the risk–coverage curves on the internal test set. Table V summarises the selective prediction results.

D. Heuristic Visible Lesion-Area Proxy

The heuristic visible lesion-area proxy was computed for all Main19 images across training, calibration, and internal test splits (total: 20,562 images, excluding healthy classes for lesion interpretation but including them in the computation pool). Fig. 5 presents a qualitative panel.

Across splits, mean visible lesion area is approximately 6.2% (calibration: 6.19%; internal test: 6.22%; training: 6.26%), with a median of approximately 4.4% and standard deviation of approximately 6.5%. The close agreement across

splits indicates that partition stratification preserved lesion-area distributions.

Notable class-level variation is observed. Classes with large proxy lesion areas include potato early blight (mean 12.47%), corn gray leaf spot (13.01%), and corn northern leaf blight (11.42%). Classes with smaller proxy areas include tomato mosaic virus (1.79%), corn healthy (0.86%), and tomato healthy (2.73%). These differences are qualitatively plausible given the known visual symptom morphology of each condition—early blight and late blight produce large necrotic lesions, whereas mosaic virus primarily causes colour mottling without prominent discrete lesion areas. However, these observations do not constitute validation against expert-annotated lesion ground truth and must not be interpreted as agronomically validated severity estimates.

TABLE V

SELECTIVE-PREDICTION RESULTS ON THE MAIN19 INTERNAL TEST SET (5,322 IMAGES). THE CONFIDENCE THRESHOLD WAS SELECTED ON CALIBRATION DATA ONLY ($\tau = 0.50$, THE MINIMUM FLOOR). COVERAGE DENOTES THE FRACTION OF TEST IMAGES ACCEPTED; RISK DENOTES THE TOP-1 ERROR RATE AMONG ACCEPTED IMAGES. CONFIDENCE: UNCALIBRATED SOFTMAX.

Model	Threshold	Coverage	Rejected	Selective Acc.	Risk
EfficientNet-B0	0.50	1.0000	0	0.9955	0.0045
EfficientNetV2-S	0.50	0.9996	2	0.9957	0.0043
ViT-B/16	0.50	0.9994	3	0.9955	0.0045

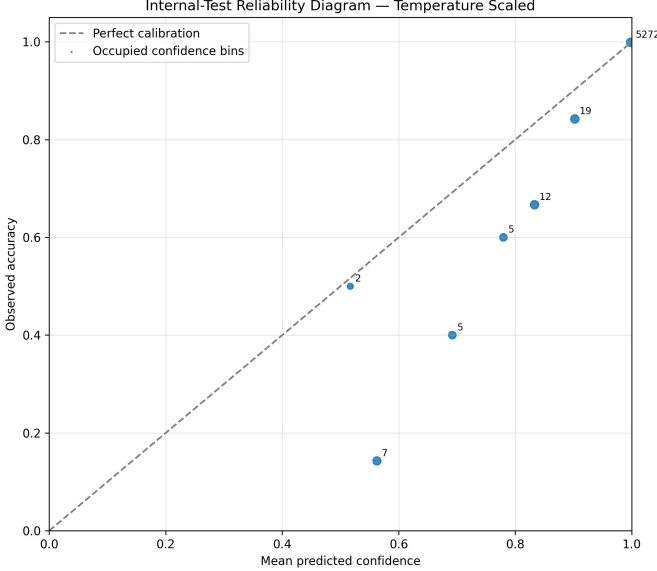


Fig. 3. EfficientNet-B0 reliability diagram on the internal test set after temperature scaling ($T = 1.5695$, 15 equal-width bins). ECE: 0.0023. The model closely tracks the identity diagonal, indicating well-calibrated confidence on the source-domain test data.

E. External Domain Shift on PlantDoc

The EfficientNet-B0 baseline was applied zero-shot to the 144-image, 15-class semantically mapped PlantDoc external test subset. The results reveal severe domain shift.

Overall metrics. Top-1 accuracy: **0.1944** (28 of 144 correct). Balanced accuracy: 0.1998. Macro-F1: 0.1428. Weighted-F1: 0.1698. Top-5 accuracy: **0.6736**.

The contrast with internal performance (99.53% top-1 accuracy) is stark: the model was trained on controlled, single-leaf images against uniform backgrounds, while PlantDoc contains field-acquired images with complex natural backgrounds, variable lighting, and co-occurring foliage.

Top-5 analysis. The high top-5 accuracy (67.4%) indicates that the correct class is retrievable within the model’s top-5 predictions for the majority of images, even when it is not the top-ranked class. This suggests that the model retains some ability to distinguish relevant disease features under domain shift, but that ranking confusion prevents reliable top-1 identification.

Confidence analysis. Despite poor accuracy, the model remains overconfident: mean confidence on all 144 Plant-

TABLE VI
PLANTDOC ZERO-SHOT EXTERNAL EVALUATION SUMMARY
(EFFICIENTNET-B0, 144 IMAGES, 15 SEMANTICALLY MAPPED CLASSES, $T=1.0$).

Metric	Value
Top-1 accuracy	0.1944
Top-5 accuracy	0.6736
Balanced accuracy	0.1998
Macro-F1	0.1428
Weighted-F1	0.1698
Mean confidence (all)	0.845
Median confidence (all)	0.942
Mean confidence (correct)	0.883
Mean confidence (incorrect)	0.836

Doc images is 0.845, with median 0.942. Correct predictions have mean confidence 0.883; incorrect predictions have mean confidence 0.836. The small gap (0.047) indicates that confidence alone is insufficient to identify domain-shifted misclassifications. When the source-domain selective threshold ($\tau = 0.50$) is applied, only 11 of 144 images are rejected (coverage: 92.4%) and retained-case accuracy improves only marginally to 19.5%. The calibration-derived threshold provides minimal protection against domain-shift errors, confirming that source-domain confidence thresholds do not provide sufficient protection under severe external shift.

Fig. 6 shows the confidence distributions for correct and incorrect predictions on PlantDoc.

Table VI summarises the overall external metrics.

Per-class and per-crop analysis. Table VII presents the per-class PlantDoc results. Six of 15 classes achieve zero correct top-1 predictions. At the crop level, bell pepper achieves the highest top-1 accuracy (0.353), followed by corn (0.269), potato (0.182), and tomato (0.139). Fig. 7 shows the external-domain confusion matrix. Fig. 8 shows top-1 accuracy by crop.

The PlantDoc result should be interpreted as evidence of substantial source-to-target domain shift within the evaluated mapped subset, rather than as a direct estimate of universal field-deployment performance across crops, regions, cultivars, devices, and agronomic conditions.

F. Retrieval-Grounded Advisory Prototype Evaluation

The prototype was functionally tested for deterministic recovery of the label-associated advisory entry and for enforcement of confidence-policy rules across all 19 supported

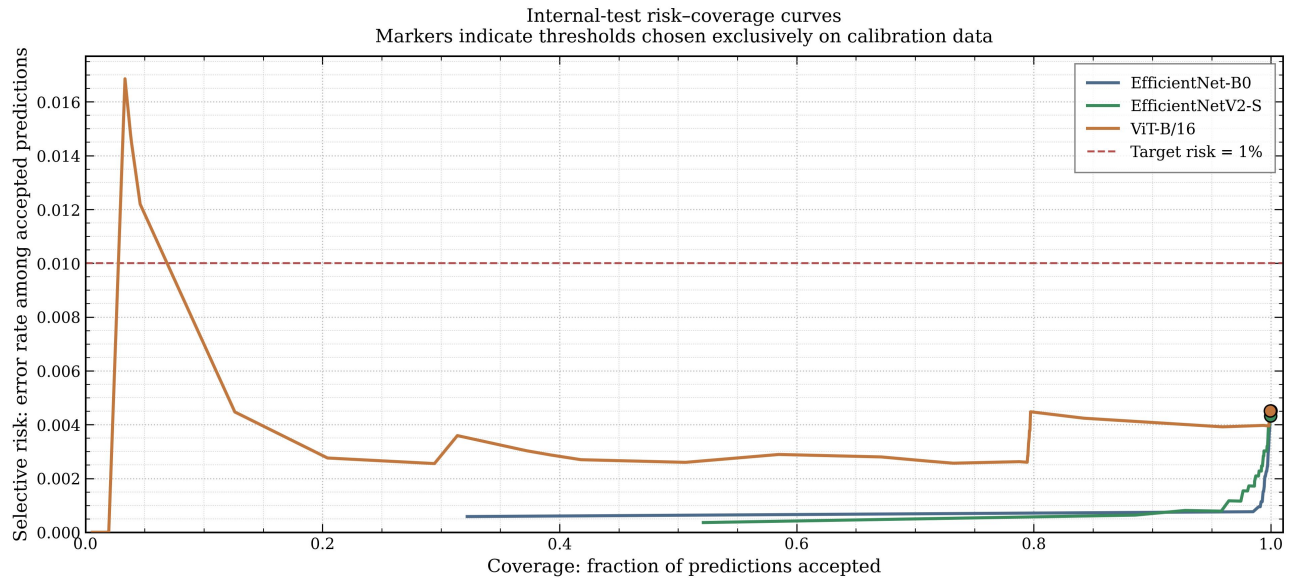


Fig. 4. Risk-coverage curves on the Main19 internal test set (5,322 images) for all three classifiers. Risk is the top-1 error rate among accepted predictions. The vertical dashed line marks the confidence threshold ($\tau=0.50$) selected using the calibration partition only. At this threshold, EfficientNet-B0 accepts all images (coverage 100%). Confidence values are uncalibrated softmax outputs.

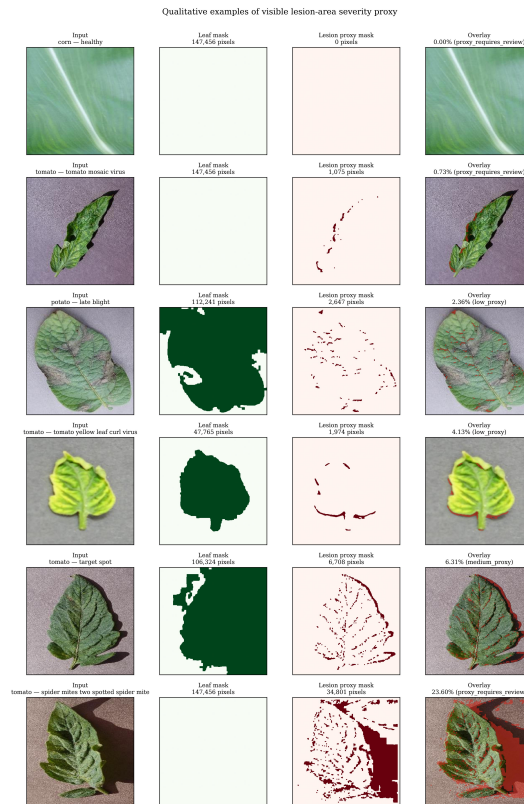


Fig. 5. Qualitative examples of the heuristic image-based visible lesion-area proxy. Highlighted pixels satisfy pre-defined lesion-like colour/texture criteria. The mild ($<5\%$), moderate (5–20%), and severe ($>20\%$) labels are proxy bands only; this procedure is not expert-annotated segmentation, a model attention map, or an agronomically validated disease-severity estimate.

labels in English, Hindi, and Telugu. All 19 labels retrieved the correct primary advisory entry (19/19 deterministic primary-entry recovery in a closed-label functional test). Advisory re-

sponses ranged from 1,568 to 1,914 characters at the medium-confidence tier, covering symptom descriptions, causal agents, and management principles.

TABLE VII
PLANTDOC-TO-MAIN19 SEMANTIC CLASS MAPPING AND PER-CLASS ZERO-SHOT RESULTS. [†]FOLDER-TO-HEALTHY MAPPINGS ARE SEMANTIC ASSUMPTIONS. SIX CLASSES ACHIEVE ZERO TOP-1 ACCURACY. TWELVE PLANTDOC FOLDERS COVERING CROPS ABSENT FROM MAIN19 WERE EXCLUDED.

PlantDoc Folder	Mapped Main19 Label	Imgs	Top-1	F1
Bell_pepper_leaf [†]	bell_pepper__healthy	8	0.750	0.240
Bell_pepper_leaf_spot	bell_pepper__bacterial_spot	9	0.000	0.000
Corn_Gray_leaf_spot	corn__cercospora_leaf_spot	4	0.500	0.250
Corn_rust_leaf	corn__common_rust	10	0.200	0.235
Corn_leaf_blight	corn__northern_leaf_blight	12	0.250	0.333
Potato_leaf_early_blight	potato__early_blight	14	0.286	0.381
Potato_leaf_late_blight	potato__late_blight	8	0.000	0.000
Tomato_leaf [†]	tomato__healthy	8	0.000	0.000
Tomato_leaf_bacterial_spot	tomato__bacterial_spot	9	0.000	0.000
Tomato_Early_blight_leaf	tomato__early_blight	9	0.111	0.133
Tomato_leaf_late_blight	tomato__late_blight	10	0.600	0.279
Tomato_mold_leaf	tomato__leaf_mold	6	0.000	0.000
Tomato_Septoria_leaf_spot	tomato__septoria_leaf_spot	12	0.167	0.211
Tomato_leaf_mosaic_virus	tomato__tomato_mosaic_virus	10	0.000	0.000
Tomato_leaf_yellow_virus	tomato__yellow_leaf_curl_virus	15	0.133	0.222
Total (15 compatible classes)		144	0.194	0.143

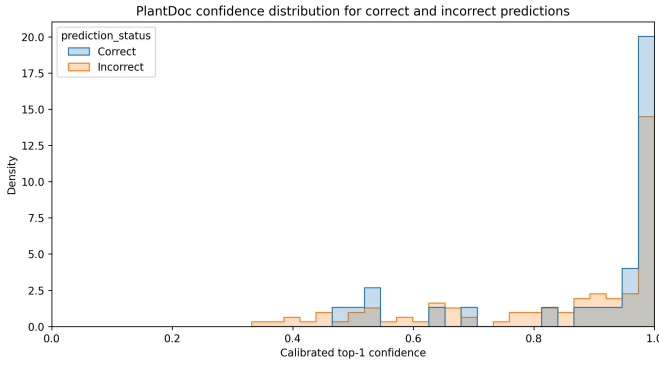


Fig. 6. Confidence distributions for correct (28) and incorrect (116) predictions on the PlantDoc external test set (144 images, EfficientNet-B0, $T = 1.0$). The distributions overlap substantially, indicating that softmax confidence does not reliably distinguish correct from incorrect predictions under domain shift. Mean confidence: correct 0.883, incorrect 0.836.

These tests verify system routing and policy behaviour; they do not constitute expert validation of agronomic advice, farmer comprehension, translation quality, or user-facing evaluation. Table VIII summarises the functional test results.

V. DISCUSSION

A. Controlled-Domain Performance in Context

The EfficientNet-B0 baseline achieves 99.53% top-1 accuracy on the locked internal test set—a result consistent with the broader literature on PlantVillage models [3], [4], [7]. However, this study deliberately places this result in context: it is a ceiling observed under matched imaging conditions, not a guarantee of field-level performance.

Under the evaluated models and split protocol, larger architectures did not show a practically meaningful internal-

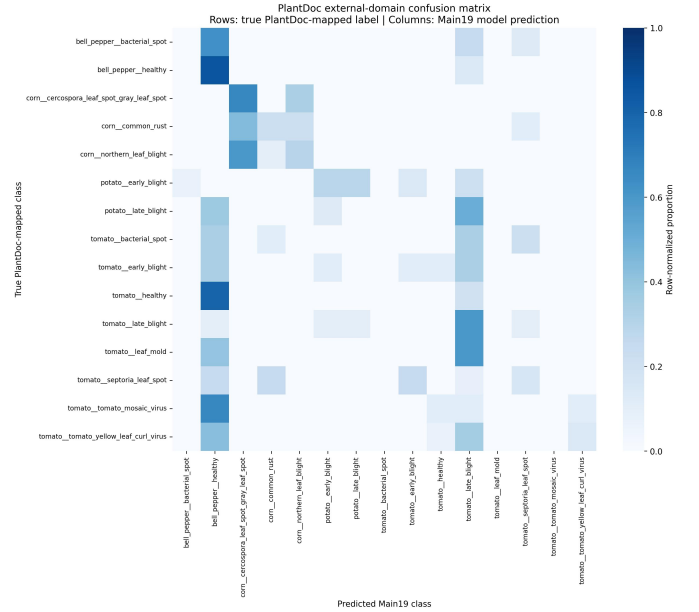


Fig. 7. Confusion matrix for EfficientNet-B0 on the semantically mapped PlantDoc external test subset (144 images). Rows: 15 PlantDoc-mapped true labels. Columns: Main19 predicted labels. The model outputs predictions across all 19 Main19 classes; 6 classes never appear as predictions on this subset. The low diagonal density confirms severe top-1 accuracy degradation under domain shift.

domain advantage over EfficientNet-B0. All three models achieved accuracy above 99.3% and macro-F1 above 0.992, with parameter counts ranging from ≈ 5.3 M (B0) to 85.8M (ViT-B/16). The near-identical performance suggests that the discriminative features required for near-perfect classification on this controlled task are accessible to all three architectures,

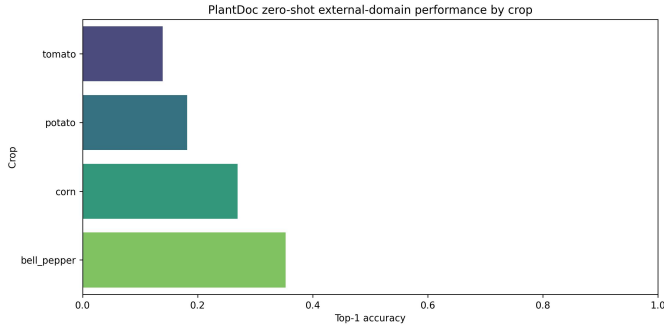


Fig. 8. PlantDoc zero-shot top-1 accuracy grouped by crop in the semantically mapped external subset. Crop-level values should be interpreted cautiously because crop groups differ in image count and number of mapped classes.

TABLE VIII
RETRIEVAL-GROUNDED ADVISORY PROTOTYPE: FUNCTIONAL TEST SUMMARY. THE 19/19 LABEL-MATCH RESULT REFLECTS DETERMINISTIC ROUTING IN A CLOSED-LABEL TEST, NOT OPEN-QUERY RETRIEVAL PERFORMANCE.

Property	Value
Knowledge base entries	19
Primary label recovery	19/19 (deterministic)
Languages	English, Hindi, Telugu
Advisory length range	1,568–1,914 chars
Retrieval method	TF-IDF cosine similarity
Pesticide prescription	Prohibited
Expert confirmation (<0.70)	Required
Field-verification (≥ 0.90)	Mandatory warning
KB entries expert-reviewed	No

and the differentiating factors become calibration quality, parameter efficiency, and generalisation under shift.

B. Calibration: Metric-Dependent Improvement

Temperature scaling with $T = 1.5695$ reduced internal-test ECE by 29% (from 0.00321 to 0.00228) and Brier score from 0.00704 to 0.00642, but NLL increased slightly from 0.01550 to 0.01593. This mixed result underscores that calibration quality depends on the specific metric used to assess it. The ECE and Brier improvements are consistent with the model’s confidence distribution being softened toward the identity diagonal on the reliability curve. The slight NLL increase may reflect the trade-off between bin-level calibration and pointwise log-likelihood.

The PlantDoc confidence analysis reveals a critical limitation of calibration under distribution shift: despite being calibrated on PlantVillage data, the model assigns high confidence (median 0.942) to most PlantDoc images, including a large majority of misclassified ones. The mean confidence gap between correct and incorrect external predictions (0.047) is negligible. This finding confirms that post-hoc calibration on source-domain data does not generalise to target-domain confidence estimates [11], [22].

C. The External Domain Shift Gap

The 80.1 percentage-point drop in top-1 accuracy between the internal test (99.53%) and the PlantDoc external test (19.44%) is the most important finding of this study. The PlantDoc subset comprises only 144 images from 15 classes (mean 9.6 per class), and several classes contain as few as 4 images. Conclusions about specific class difficulty must therefore be made cautiously.

Nevertheless, the result is consistent with published external evaluations of PlantVillage-trained models on PlantDoc [10] and reinforces that the controlled-domain accuracy advantage does not transfer reliably. The top-5 accuracy of 67.4% suggests that the model has not lost all discriminative knowledge: the correct answer is within the model’s top five candidates for most images. This provides a basis for designing human-in-the-loop systems that present multiple candidate diagnoses rather than forcing a single output.

When transferred to PlantDoc, the source-domain threshold ($\tau = 0.50$) rejected only 11 of 144 images and improved retained-case accuracy only marginally, indicating that source-domain confidence thresholds did not provide sufficient protection under severe external shift.

D. Lesion Proxy Limitations

The heuristic visible lesion-area proxy provides a lightweight image-characterisation mechanism that does not require a separate segmentation model. The class-level patterns are qualitatively plausible given known visual symptom morphology. However, the proxy has not been validated against expert-annotated lesion masks, and its heuristic colour/texture rules may confound image artefacts, natural leaf variation, or reflection with pathological tissue. The proxy should be treated as an exploratory tool for qualitative analysis and future annotation prioritisation, not as a disease severity estimator. It does not identify causal model features and does not constitute classifier-attention analysis.

E. Advisory Prototype and Responsible Deployment

The retrieval-grounded advisory prototype addresses the gap between classification output and usable field guidance. By conditioning advisory content on classifier confidence and explicitly enforcing a no-pesticide-prescription policy, the system avoids the most dangerous overclaims possible from an automated diagnosis system. The three-language support (English, Hindi, Telugu) begins to address the multilingual needs of the primary intended user base.

However, the knowledge base was authored for this project and has not been reviewed or endorsed by professional plant pathologists. TF-IDF retrieval is sensitive to vocabulary and lacks semantic understanding. The 19/19 primary-entry recovery result reflects deterministic routing in a closed-label test, not competitive retrieval over a large corpus. A rigorous evaluation would require expert review of advisory quality, field user studies, translation quality assessment, and adversarial testing with out-of-domain inputs.

F. Limitations

Controlled source data. PlantVillage images were collected under laboratory conditions. The demonstrated external performance (19.44% top-1 on PlantDoc) confirms that this fundamentally limits generalisation.

Non-exhaustive near-duplicate audit. Of 136 pHash near-duplicate candidates, only 2 were reviewed and confirmed as leakage; 134 remain unreviewed. The final partitions are therefore not guaranteed to be completely free of near-duplicate cross-split contamination.

PlantDoc mapping assumptions. Mapping PlantDoc folder names to Main19 classes required semantic interpretation. The mappings of `Tomato_leaf` to `tomato__healthy` and `Bell_pepper_leaf` to `bell_pepper__healthy` are assumptions. Mapping errors would affect class-level metrics.

PlantDoc subset size. With only 144 images across 15 classes (mean 9.6 per class), class-level PlantDoc metrics have high variance and should be interpreted with caution.

Cross-notebook metric discrepancy. EfficientNet-B0 internal-test accuracy differs by one prediction between Notebook 04 (0.9953) and Notebook 07 (0.9955), and the Brier scores differ by a factor of 20, indicating different computation implementations. Cross-model Brier comparisons are therefore not reported.

Missing V2-S temperature-scaled test metrics. Temperature-scaled internal-test evaluation was conducted only for EfficientNet-B0. No temperature-scaled cross-model comparison is possible.

Proxy validation. The visible lesion-area proxy has not been validated against expert-annotated ground truth. All proxy band assignments are heuristic.

Advisory evaluation depth. The advisory prototype was tested only for structural correctness (label routing and policy enforcement). Agronomic quality, user comprehension, translation accuracy, and safety under adversarial inputs have not been evaluated.

No statistical significance tests. No confidence intervals or significance tests were computed. Differences between closely performing models should not be interpreted as statistically meaningful.

VI. CONCLUSION

This paper presents a reproducible, leakage-aware crop disease classification pipeline evaluated across controlled and external domains. The principal finding is that high controlled-domain accuracy—here 99.53% for EfficientNet-B0 on Main19—does not transfer to field-like imagery: zero-shot evaluation on a semantically mapped PlantDoc subset yields 19.44% top-1 accuracy despite a top-5 accuracy of 67.4%. Strong controlled-domain performance did not transfer reliably to the semantically mapped PlantDoc external subset, demonstrating that high internal accuracy alone is insufficient evidence of robustness to field-like imaging conditions.

Post-hoc temperature scaling ($T = 1.5695$) reduced ECE from 0.00321 to 0.00228 and Brier score from 0.00704 to 0.00642 on the internal test set, while NLL increased

slightly from 0.01550 to 0.01593; calibration improvement was metric-dependent. Selective prediction with a calibration-derived threshold of 0.50 achieved full coverage at sub-0.5% risk internally, but provided only marginal protection against domain-shift errors externally. The heuristic visible lesion-area proxy revealed qualitatively plausible class-level variation in visible lesion area, while its limitations as an unvalidated heuristic are explicitly documented.

The retrieval-grounded multilingual advisory prototype was functionally tested for deterministic label-associated entry recovery and confidence-policy enforcement. It was not evaluated for expert agronomic correctness, farmer comprehension, or safety under adversarial and out-of-domain inputs.

Future directions include: domain adaptation pre-training on unlabelled field imagery; collection of expert-validated lesion annotations to upgrade the lesion proxy; expansion of the PlantDoc evaluation to more crops and regions; completion of the near-duplicate audit for remaining unreviewed candidates; and rigorous field user studies of the advisory system with professional agronomists.

DECLARATIONS

Data availability. The PlantVillage dataset is publicly available at <https://plantvillage.psu.edu/>. PlantDoc is available at <https://github.com/pratikkayal/PlantDoc-Dataset>. The derived split manifests, class maps, and evaluation metadata are available in the project repository.

Code availability. All analysis notebooks and reproducibility artefacts are available in the project repository at random seed 7 with Python 3.12.13 / PyTorch 2.10.0 / torchvision 0.25.0.

Conflicts of interest. The authors declare no conflicts of interest.

Responsible-use statement. The advisory prototype described in this paper is designed for educational and screening purposes only. It does not constitute a confirmed agronomic or clinical diagnosis and must not be used to prescribe pesticides, chemicals, or treatments without professional agronomic review.

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